UNIVERSITY OF PISA

Lesson 11

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From statistical test to gene list

Things to do before biology:

- **Split up- and down-regulated genes**
  - e.g. padj < 0.05 and |log2FC| > 1
- **Check for obvious artifacts**
  - genes with very low counts in all samples (it should be not necessary if you cleaned the data before DE)
  - mitochondrial / ribosomal genes dominating the list
- **Map IDs**
  - from Ensembl → Gene Symbol / Entrez, etc. (needed by many enrichment tools)

Map IDs

- R / Bioconductor (most common in RNA-seq)
  - Biomart
    - Query Ensembl's BioMart from R.
    - Convert between: Ensembl gene/transcript IDs, gene symbols, Entrez IDs, UniProt, etc.
    - Great for: bulk conversion of thousands of IDs in scripts.
  - AnnotationDbi + org.*.db packages
    - e.g. org.Hs.eg.db, org.Mm.eg.db.
    - Local SQLite databases;
    - very fast, no internet needed.
    - Good for: Entrez ↔ symbol ↔ other IDs in standard Web-based tools organisms.

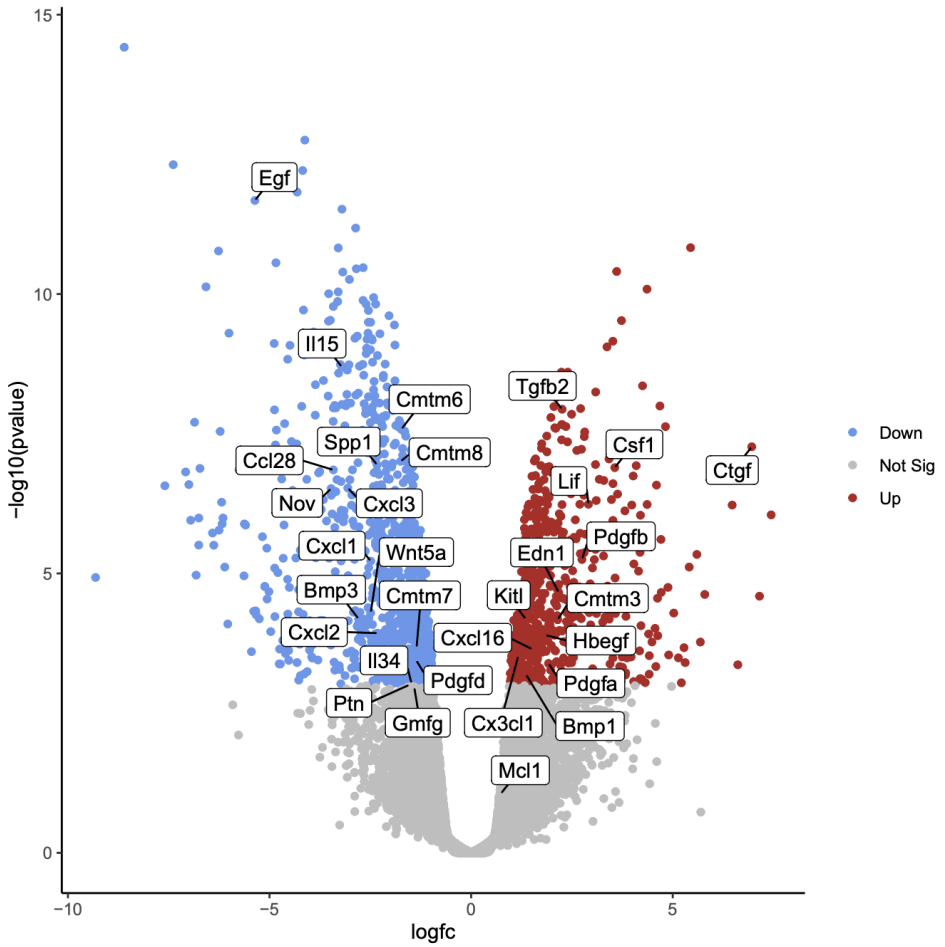
Map IDs

- Web-based toolsBiomart
  - Ensembl BioMart web interface
    - Point-and-click version of biomaRt.
    - You can upload a list of Ensembl IDs and download a table with symbols, Entrez, UniProt, etc.
  - g:Profiler (g:Convert)
    - Web tool (and API/R package) for ID conversion across many databases and species.
    - Handles mixed ID types reasonably well.
    - Also integrates directly with GO / pathway enrichment.

How do we look at the most significant genes?

Volcano plot

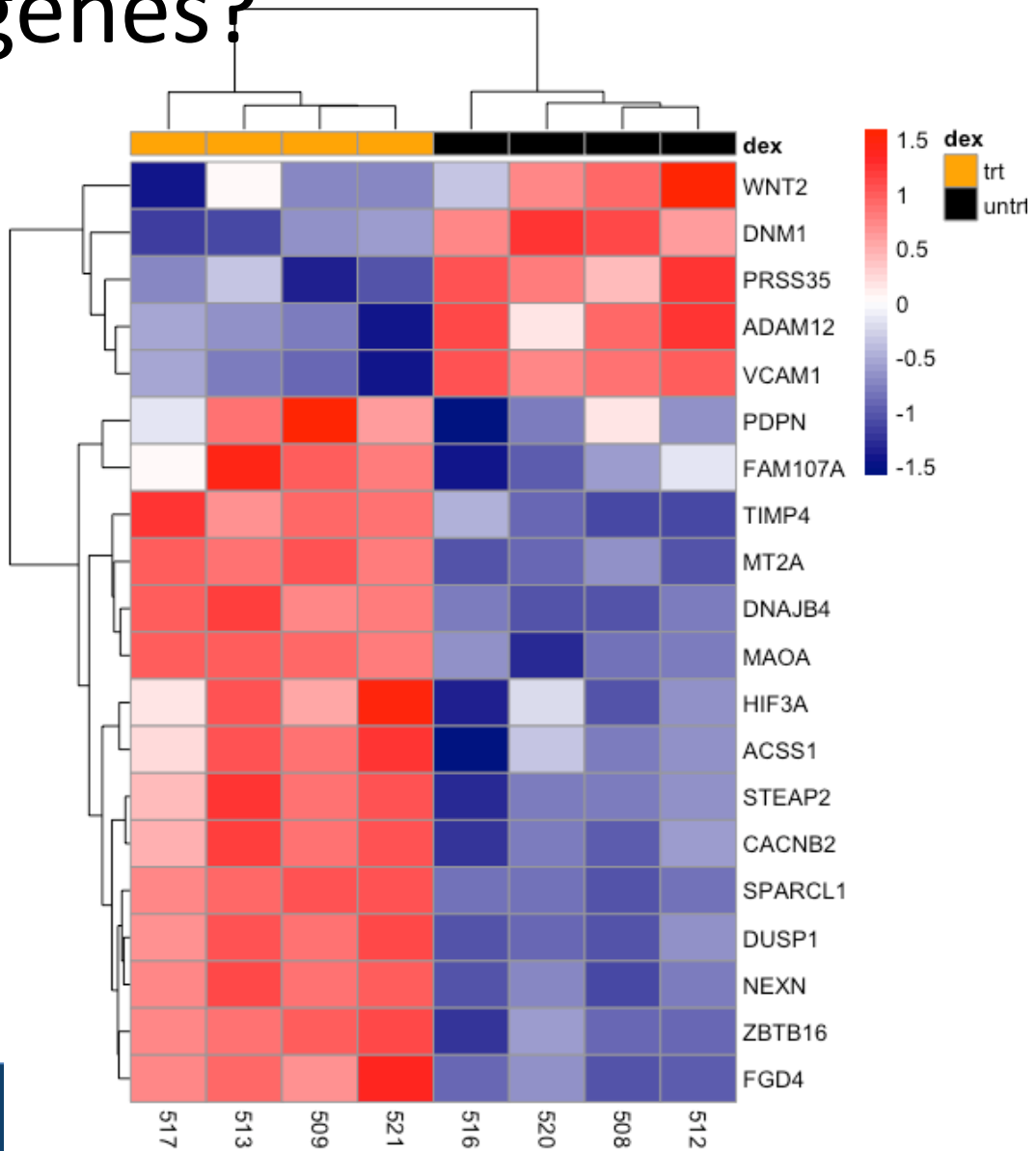
- x: log2 fold change,
- y: -log10(adjusted p-value)
- Highlight interesting genes as examples.



How do we look at the most significant genes?

Heatmap of top N genes

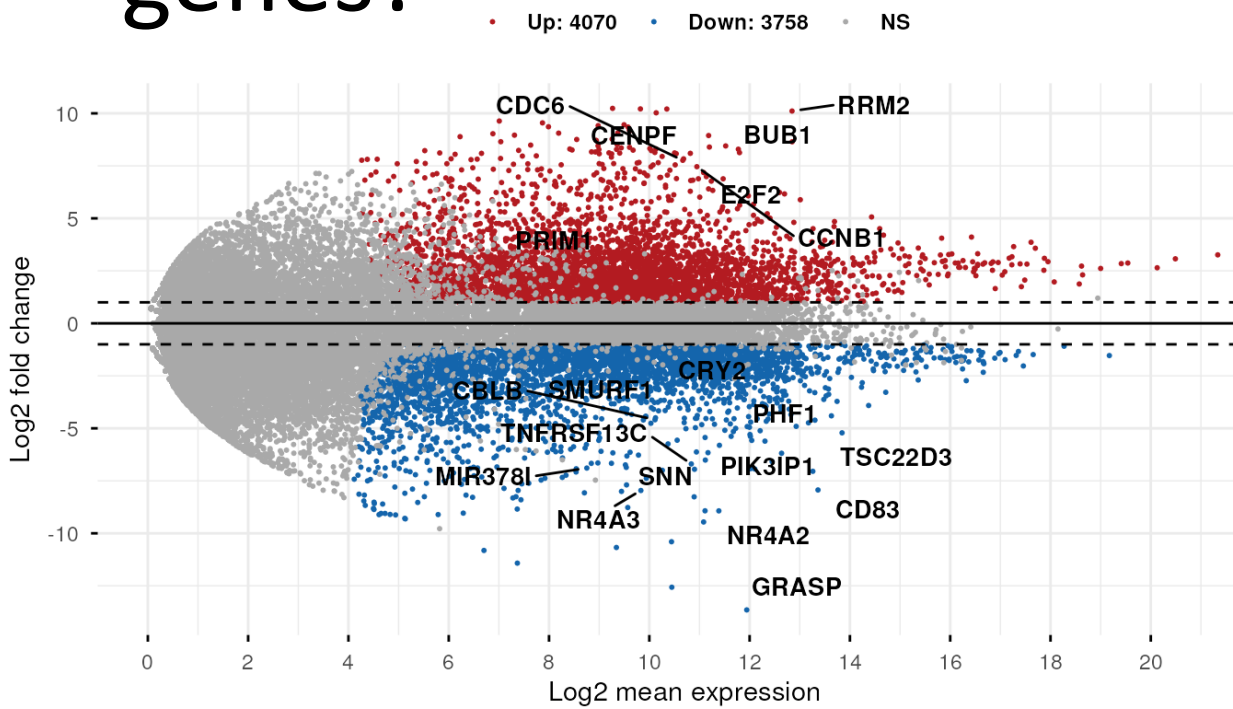
- Use normalized values (e.g. logCPM / vst).
- Samples in columns, genes in rows.
- Cluster to see if samples separate as expected.



How do we look at the most significant genes?

MA plot useful for

- Check symmetry of changes
  - Under a well-normalized comparison, most genes should be centered around M ≈ 0.
  - If the cloud is shifted up or down → possible normalization problem or strong global shift.

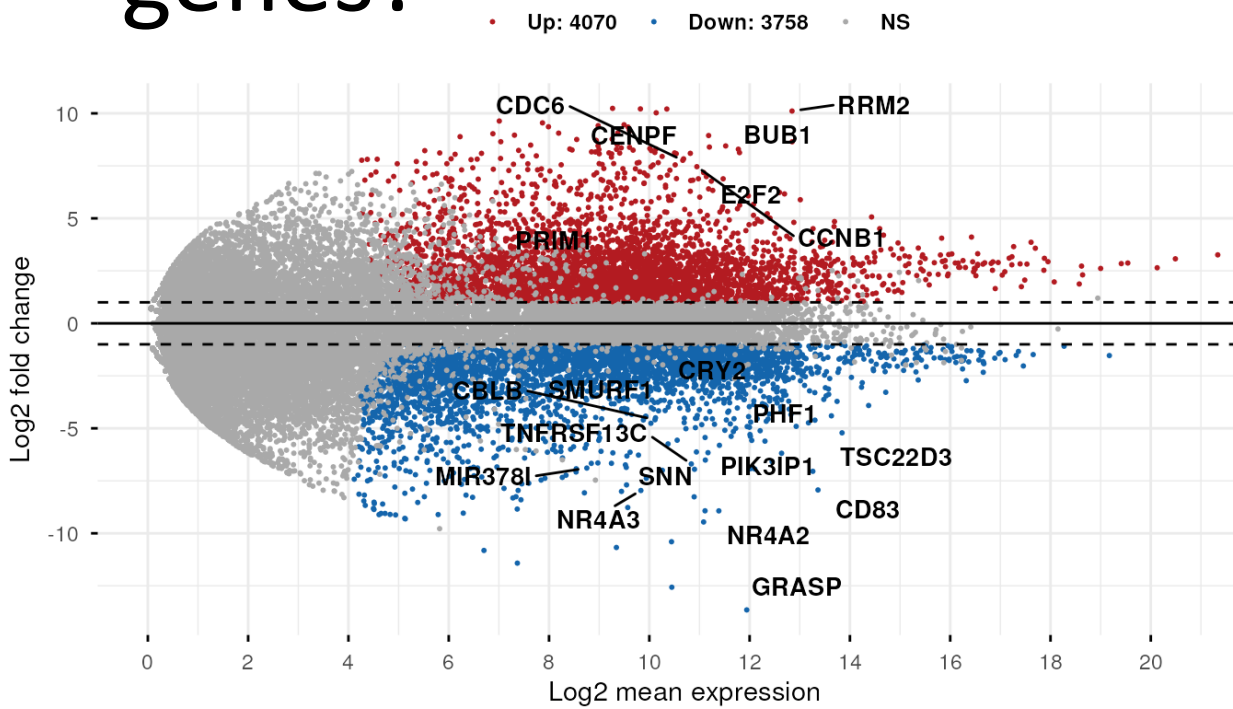


- See how effect size depends on expression level
  - At very low A (low counts), you expect more noise and more variable M.
  - At high A, estimates should be more stable.
  - You can quickly see if large fold-changes are mostly coming from low-expression genes (often less reliable).

How do we look at the most significant genes?

MA plot useful for

- Spot weird patterns or artifacts
  - Curved trends (e.g. M increasing with A) can indicate: compositional or library-size bias not fully corrected, batch or GC-content issues, other systematic effects.
  - Clusters or lines of genes may flag specific gene families or technical artifacts.



- Visualize DE genes in context
  - Coloring significant DE genes on the MA-plot shows:
    - whether they're spread across the expression range,
    - or mostly at high/low expression.

Unranked or ranked

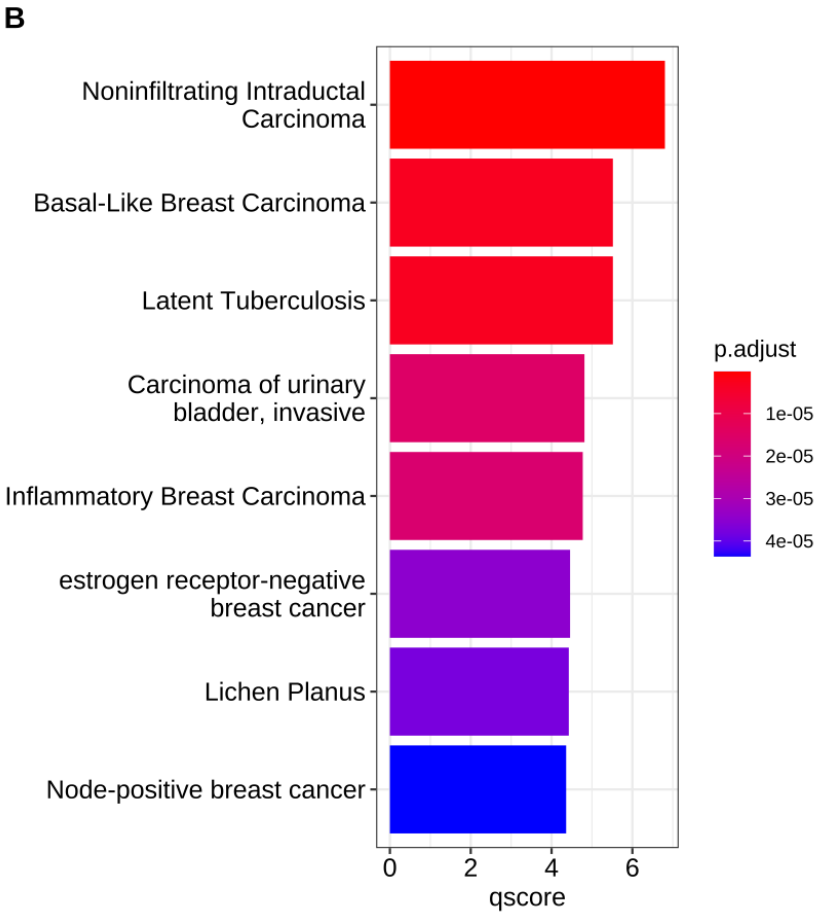
- Gene list can be:
  - Unranked (e.g. genes with somatic mutations in cancer sample)
  - Ranked (e.g. sensitivity in a CRISPR screen)
- RNA-seq differential expression analysis produces ranked lists
- Ranked lists are ordered by a score or metric:
  - e.g. adjusted p-value
  - e.g. log2fold change
- Ranked lists can also have a threshold applied:
  - e.g. adjusted p-value < 0.05

Genes	|log (fold change)|	False discovery rate (FDR)
HOXB7	2.085898	1.33E−07
RHPN2	2.147406	6.32E−07
SRGN	5.174341	3.34E−06
FOXF2	2.318869	4.51E−06
PLVAP	3.404251	1.98E−05
COX7A1	2.988735	5.41E−05
APOE	3.399852	6.21E−05
SPINT2	2.301193	6.65E−05
LXN	2.523381	8.93E−05
TNFRSF1B	2.427194	1.10E−04
VAMP8	3.494650	1.61E−04
CBS	2.868625	1.76E−04
HCLS1	2.686705	1.81E−04
PHGDH	2.362045	2.15E−04
GIMAP7	2.258177	2.74E−04
C1QC	2.885076	2.76E−04
HBB	5.191561	2.86E−04
C1QA	3.521570	2.93E−04
TYROBP	3.511873	3.17E−04
C1QB	3.184835	3.17E−04

From EBI course: “Introduction to RNA-seq and functional interpretation”

Functional enrichment analysis

- It is a computational method used to identify biological functions, pathways, or annotations that are significantly over-represented in a list of genes
- Systematically relates your data to existing knowledge
- Can help you to:
  - Gain biological insight
  - Generate new hypotheses
  - Validate your experiment



Over-Representation Analysis (ORA)

Input: list of “significant” genes (top DE genes).

Gene Ontology (GO) enrichment

- Biological Process (BP) for processes
- Cellular Component (CC)
- Molecular Function (MF)

Pathway enrichment

- KEGG, Reactome, WikiPathways

Statistics:

- typically **hypergeometric / Fisher’s exact test**
- also apply FDR correction here!

Gene Ontology (GO)

What is GO?

- A **controlled vocabulary** to describe what genes/proteins *do*.
- Same terms used across species → enables cross-species comparison.

Three main branches

- **Biological Process (BP)** – *"What larger process is this gene involved in?"*
  e.g. cell cycle, apoptosis, immune response
- **Molecular Function (MF)** – *"What does this gene product do at the molecular level?"*
  e.g. kinase activity, DNA binding
- **Cellular Component (CC)** – *"Where in the cell is it active?"*
  e.g. nucleus, mitochondrion, plasma membrane

Structure

- Organized as a **directed acyclic graph (DAG)**
  - Terms have **parent / child** relationships (is-a, part-of).
  - A gene can be annotated with **multiple** GO terms at different levels of specificity.
- Manually annotated by a large consortium
- Data come from experimental and computational analyses

More genes

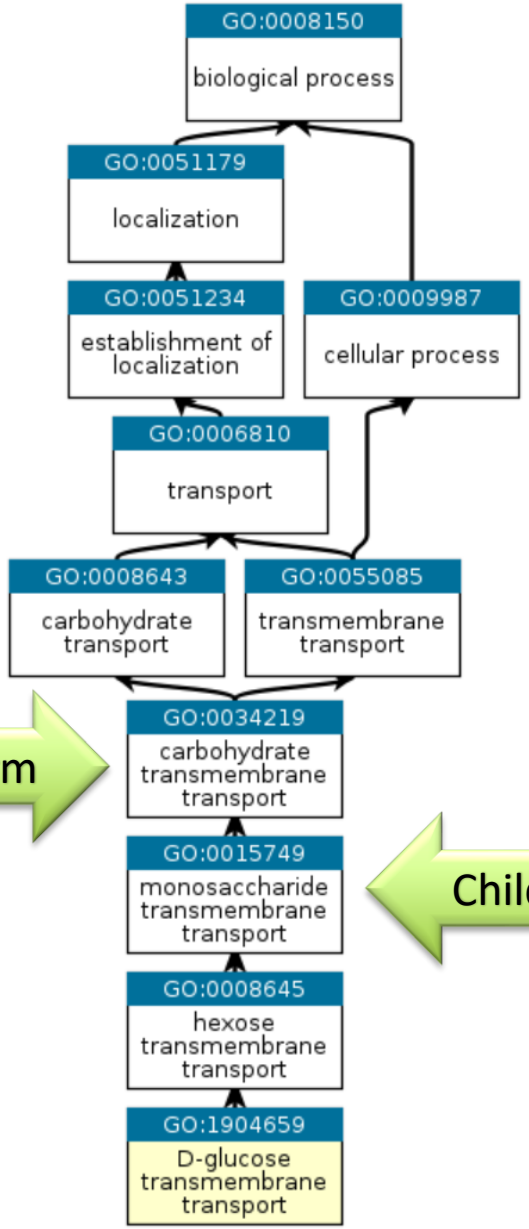
Few genes

Broad terms

Specific terms

Parent term

Child term



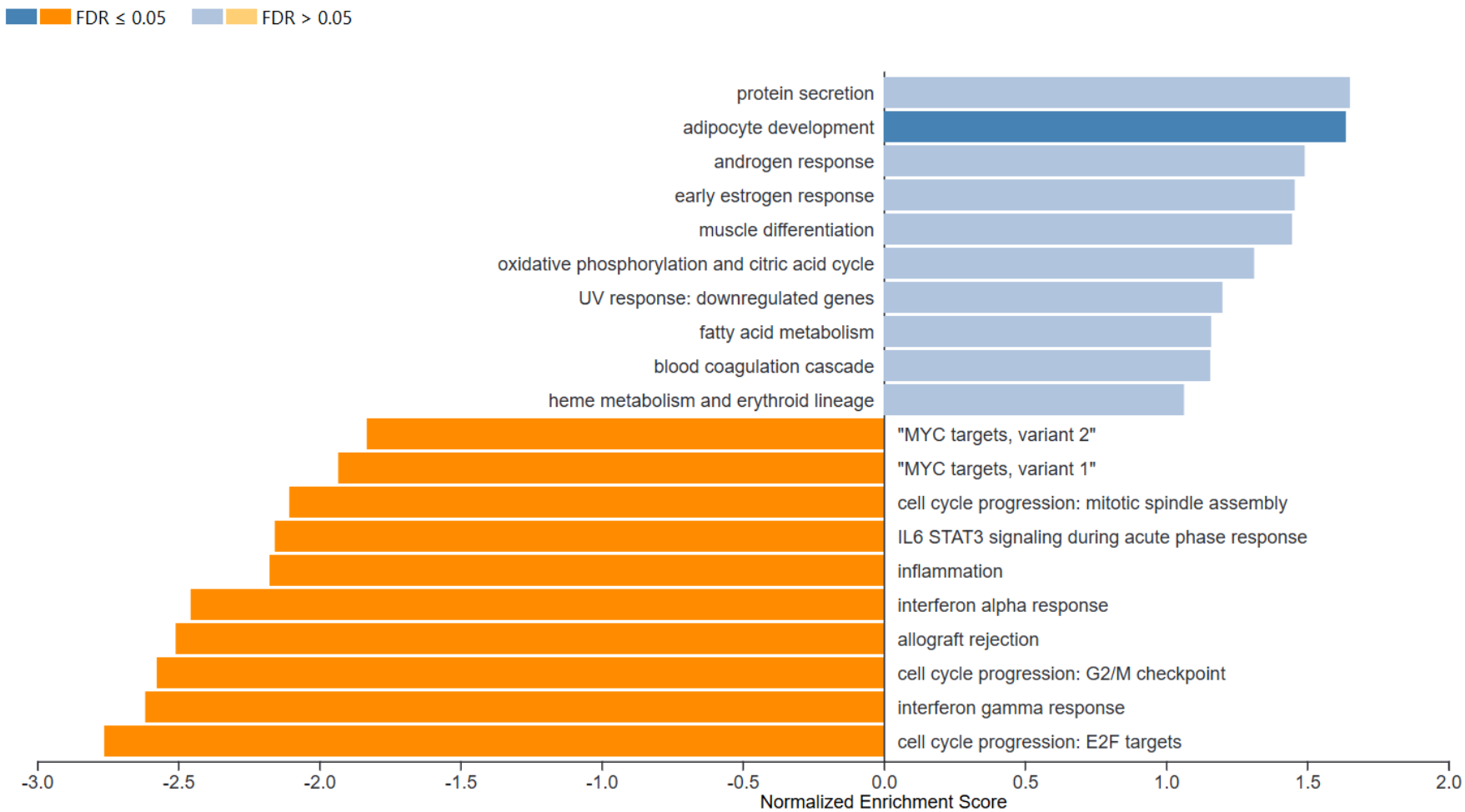
Background gene set

- Important to choose appropriate background gene set
- Wijesooriya et al. (2022) found that only 4% of papers used an appropriate background (although most failed to specify what background was used)
- Best to choose all genes that could have been captured in your experiment
- Examples:
  - All genes
  - All genes with non-zero total read count in DESeq2
  - All genes that pass DESeq2 independent filtering
  - All genes expressed in a particular tissue
  - All genes with annotations

GSEA and threshold-free gene set analysis

- Gene Set Enrichment Analysis
- GSEA: use all genes, not just a cutoff
- When you don't want an arbitrary p-adj threshold:
  - Rank genes by a **statistic** (log2FC, Wald statistic, etc.).
  - Apply **GSEA** or similar (fgsea, preranked GSEA):
  - tests whether genes in a pathway are **concentrated at the top or bottom** of the ranked list.
- Advantages:
  - uses **all** genes
  - more robust when effects are widespread but modest (few strongly DE, many weakly DE)

GSEA and threshold-free gene set analysis



Disadvantages of functional enrichment analysis

- Terms or pathways with few genes are unlikely to ever be enriched
- Hypergeometric test (the normally used as statistical test) is more likely to identify larger functional gene sets (e.g. pathways with many genes) as significant
- Genes with multiple functions can lead to enrichment of multiple terms/pathways, some of which aren't relevant
- Databases are (obviously) biased towards genes with annotation so unannotated genes (e.g. many non-coding RNA genes) are invisible to functional enrichment analysis

Functional enrichment tools

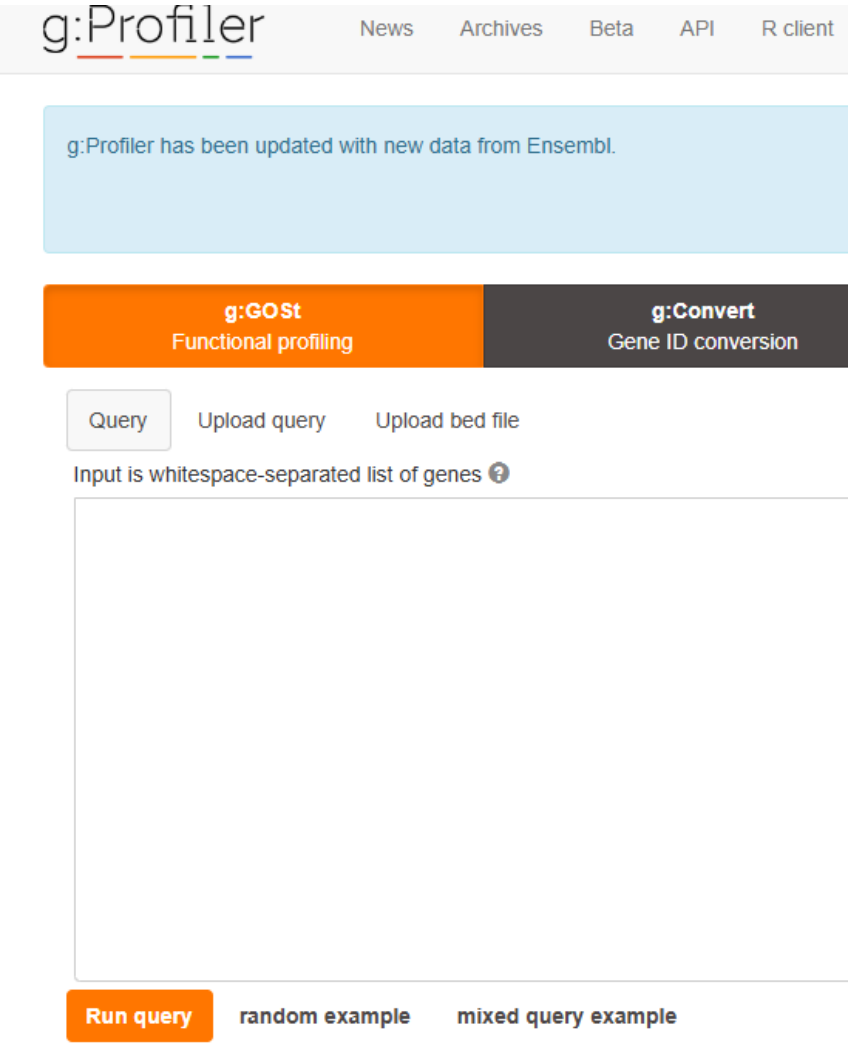
- Many, many functional enrichment analysis tools exist
- Many are created, published and then never updated
- Best to choose a well used tool

Functional enrichment tools

- Some tools:
  - **clusterProfiler (R / Bioconductor)**
    - One of the most used all-rounder for ORA and GSEA
    - Supports GO, KEGG, Reactome, MSigDB, custom gene sets.
    - Nice plotting functions (dotplot, emapplot, etc.).
    - Integrates smoothly with DESeq2/edgeR outputs.

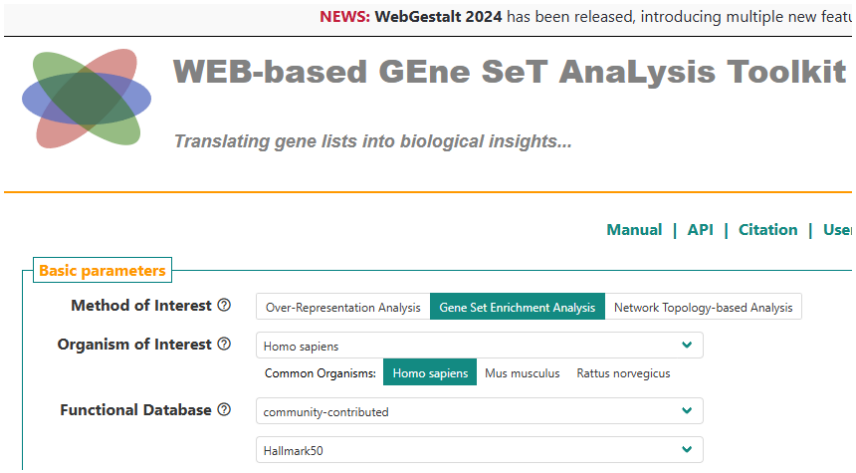
Functional enrichment tools

- Some tools:
  - **g:Profiler (g:Convert / g:GOSt)**
    - **Great web tool + R/Python API**
    - Web interface for quick analyses, plus gprofiler2 (R) and Python client.
    - Handles ID conversion + ORA in many databases (GO, KEGG, Reactome, WP, TRANSFAC, etc.).



Functional enrichment tools

- Some tools:
  - [WebGestalt]
    - **W**EB-based **GE**ne **S**eT **A**naLysis Toolkit.
    - Web server (plus R package) for functional enrichment of gene lists.
  - **Main features**
    - Supports **multiple enrichment methods** in one place:
      - **ORA** (over-representation analysis)
      - **GSEA**
      - **NTA** (network topology–based analysis)



Functional enrichment tools

- **Enrichr**

  Web-based **gene list enrichment** tool with a huge collection of gene-set libraries.

  – Focused on **ORA-style** enrichment of gene lists (not full GSEA on ranked lists).

  – Very rich libraries:

    - Transcription factor targets, pathways, PPIs, ontologies, disease signatures, drug-perturbation signatures, cell-type markers, etc.

  – Multiple visualization options:

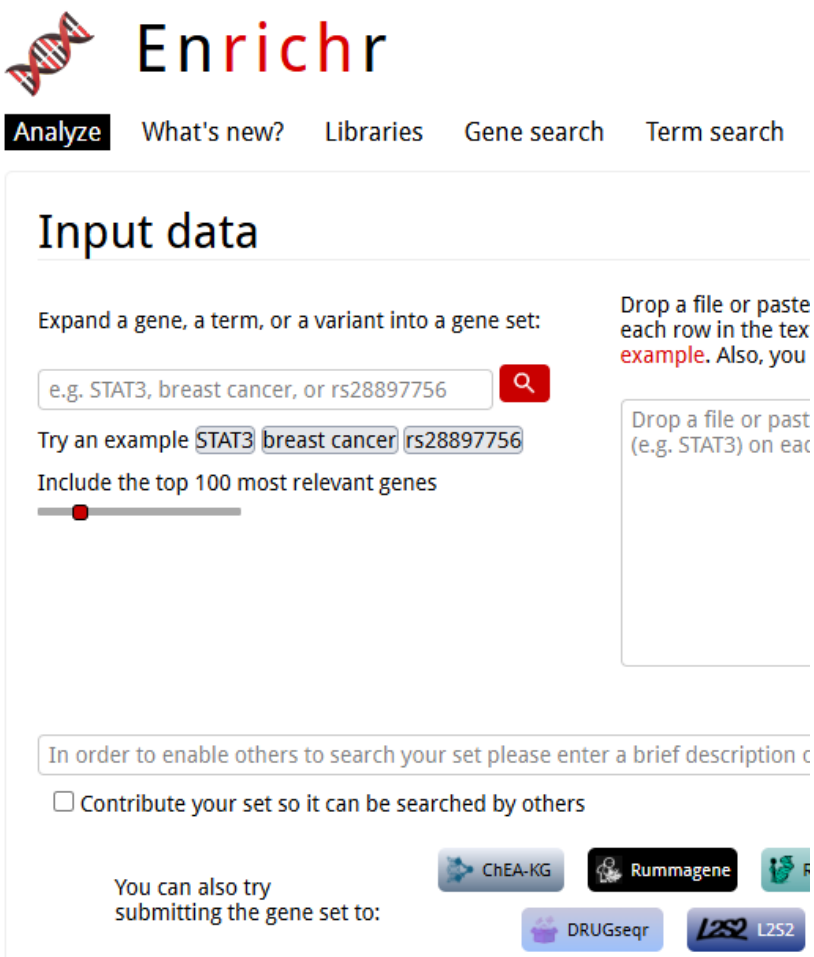
    - Bar charts, combined score rankings, network views, etc.

  – Has programmatic access:

    - **R package enrichR** and web API for integration into pipelines.

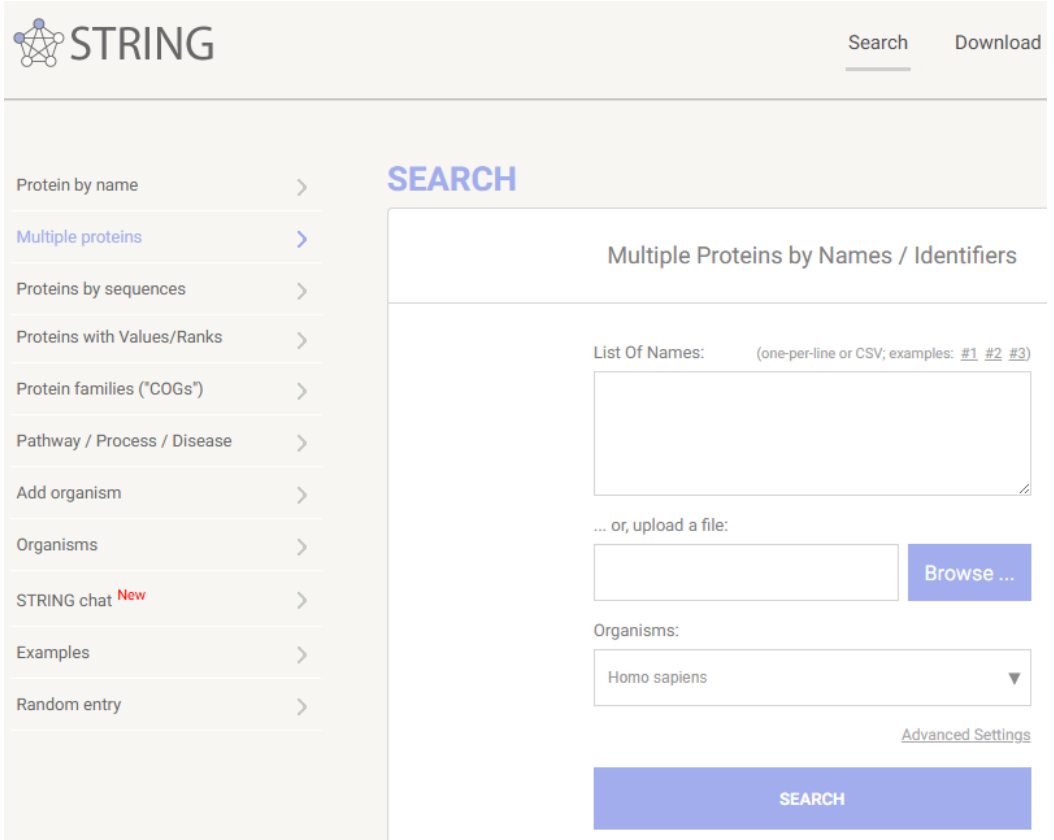
- **Good for**

  – Very quick, exploratory enrichment of a gene list across **many different library types** (especially TFs, drugs, phenotypes).



Not specific but …

- STRING – protein–protein interaction networks
  - Search Tool for the Retrieval of Interacting Genes/Proteins.
  - Online database and web tool for exploring protein–protein interaction (PPI) networks across many species.
  - Built-in functional enrichment:
    - GO, KEGG, Reactome, etc., on the network nodes.
  - Export networks (tab, images) for Cytoscape or further analysis.



Summarising functional enrichment

- Functional enrichment analysis (hopefully) summarises gene list into something shorter and more comprehensible
- But what if the list of functional enrichments is also long and/or repetitive?

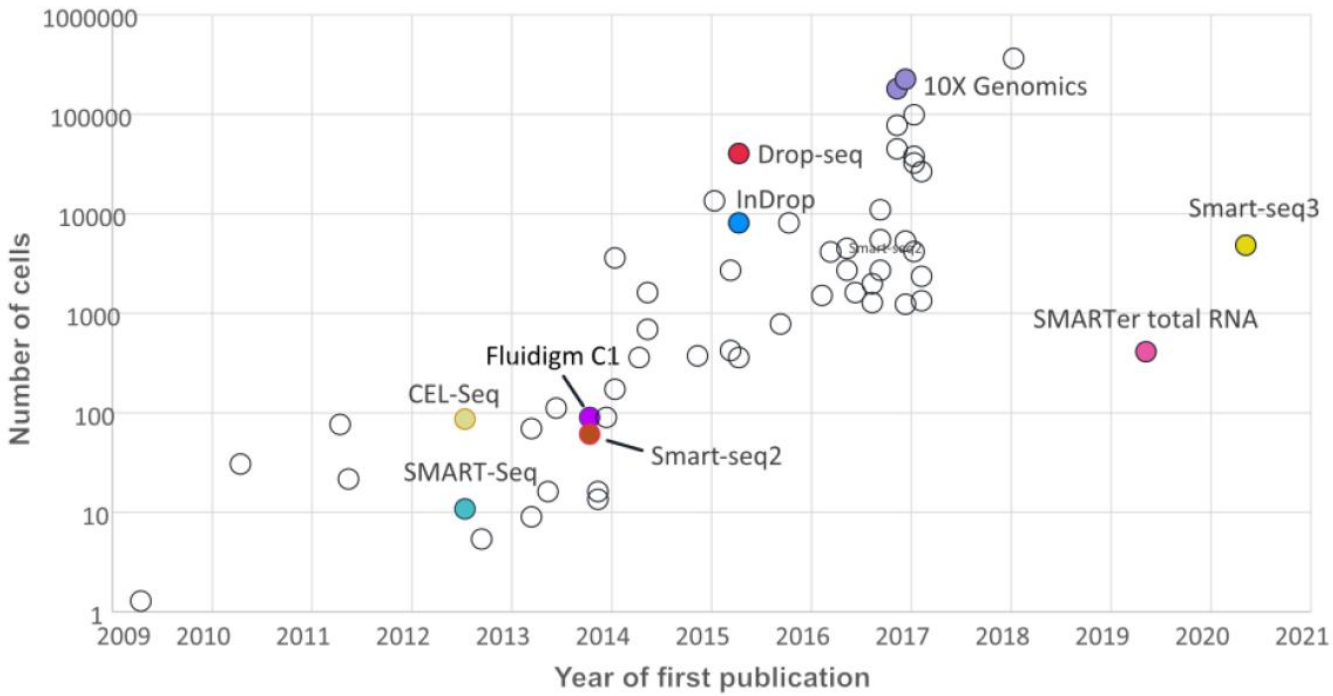
- "toolbox" :
  - Quick web
    - summaries:REVIGO
  - R/scriptable:
    - clusterProfiler simplify(), simplifyEnrichment (simplifyGO()), rrvgo
  - Network visualization:
    - Cytoscape + EnrichmentMap

Very brief introduction

SINGLE CELL RNA-SEQ

History

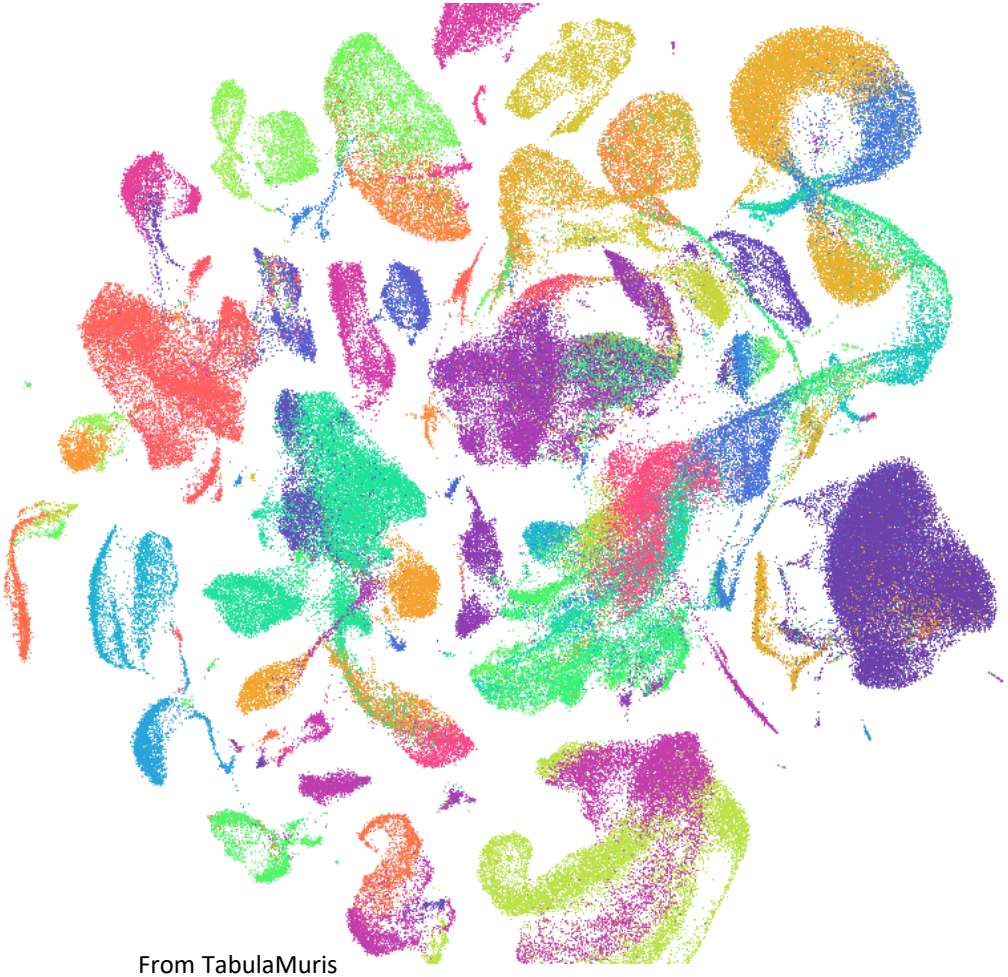
- Taeg et al. 2009
- Early developmental stage embryo: just a few cells
- Number of publications last year: around 50.000
- Now more than 50 different methods



From (Svensson, et al., 2018) with some integration and modification. A plot of some important scRNA-seq experiments reporting the analysed cell number and the year.

Cell atlases

- Now there is a great effort to construct different comprehensive single-cell atlases
- Examples:
  - Human cell atlas https://www.humancellatlas.org/
  - Mouse Tabula Muris https://tabula-muris-senis.ds.czbiohub.org/
  - Mouse cell atlas: >520,000 single cells from >10 Mouse tissues - http://bis.zju.edu.cn/MCA/index.html
  - Mouse brain: devel, adolescent, adult http://mousebrain.org/
  - Drosophila Atlas: https://flycellatlas.org/
  - … many more for tissues



From TabulaMuris

Why single cell?

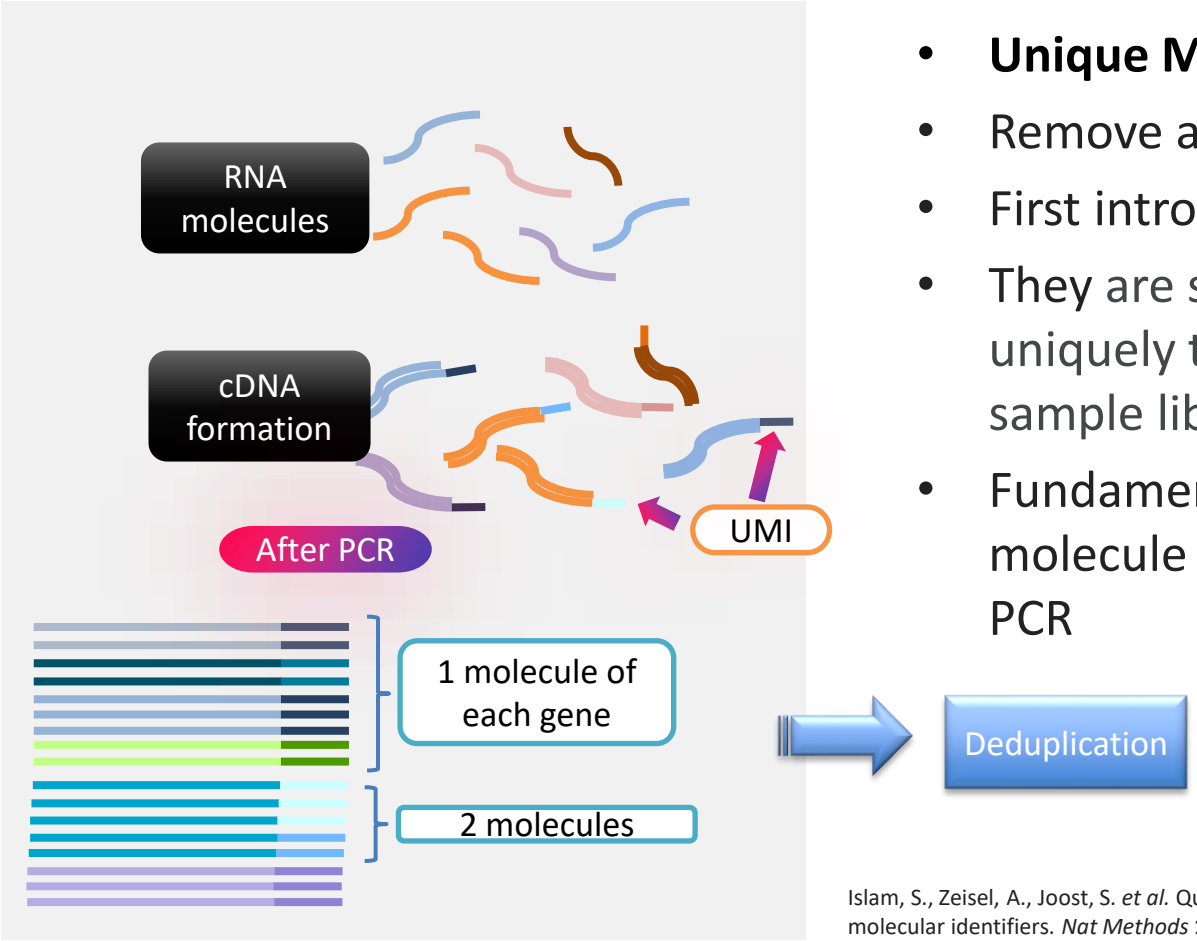
scRNA-seq

- Average information
- Information for rare cell types lost
- Easier to obtain
- Cheaper
- Quite standard workflow

Bulk RNA-seq

- Precise single-cell information
- Identification of rare cell types
- More difficult
- More expensive
- Many methods
- No standard workflow

UMI



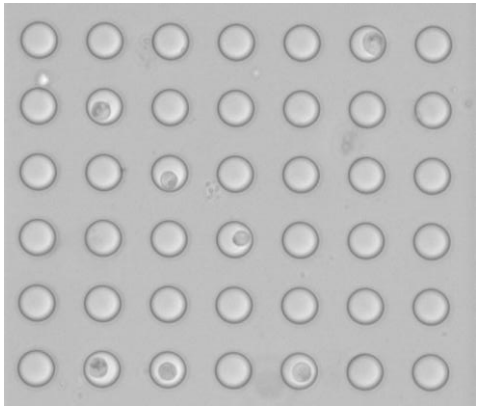
- **Unique Molecular Identifier**
- Remove amplification biases
- First introduced in 2014
- They are short sequences used to uniquely tag each molecule in a sample library
- Fundamental to count each molecule once without bias from PCR

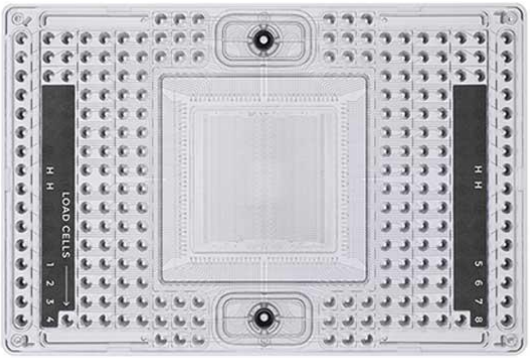
Islam, S., Zeisel, A., Joost, S. *et al.* Quantitative single-cell RNA-seq with unique molecular identifiers. *Nat Methods* **11,** 163–166 (2014).

Advantage: ease the control of the reactions

Drawback: small amount of cells

In wells methods

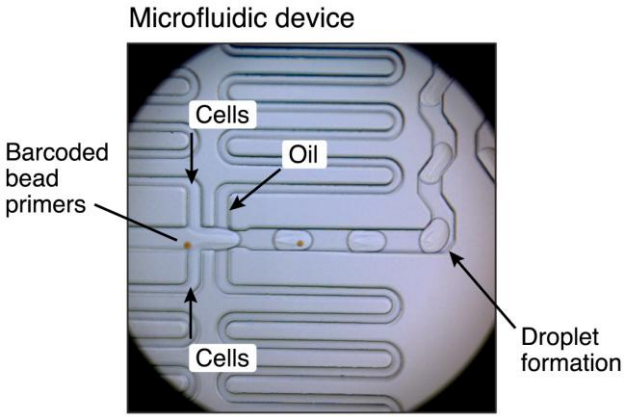


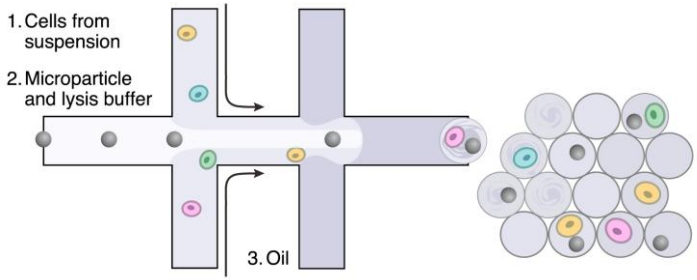


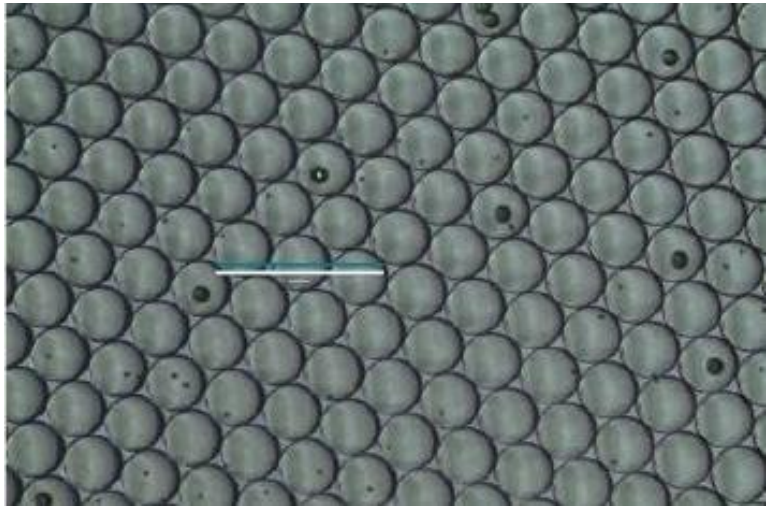
Advantages: high number of cells, not expensive

Drawback: not full length, poly A

Droplet-based methods





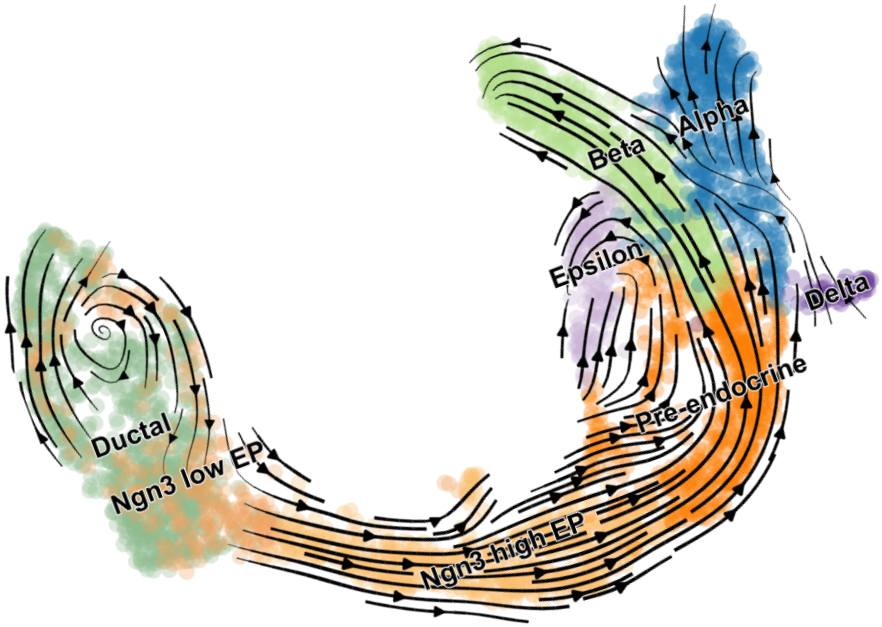


Methods comparative chart

Method	Type	Cell number	RNA full lenght	UMI	RNA species
Fluidigm C1	in wells	low throughput	yes	no	poly-A
SMARTer total RNA	in wells	low throughput	yes	no	total
Smart-seq3	in wells	low throughput	yes	yes	poly-A
SPLiT-seq	in wells	high throughput	no	yes	poly-A
InDrop	dropplets	high throughput	no	yes	poly-A
DropSeq	dropplets	high throughput	no	yes	poly-A
10X	dropplets	high throughput	no	yes	poly-A

Transcriptome Analysis Tools

- General purpose:
  - Seurat (R)
  - ScanPy (Python)
  - SingleCellExperiment (R)
  - Monocle (R)
  - scVi (Python)
  - COTAN (*Unipi*) (R)
- For pseudo-timing
  - Monocle (R)
  - Slingshot (**R**)
  - Scanpy + diffusion pseudotime & PAGA (Python)
  - Palantir (**Python**)
- For RNA stability/turnover
  - scVelo (RNA velocity–based pseudotime) (**Python**)
  - CellRank (**Python**)



OTHER OMICS

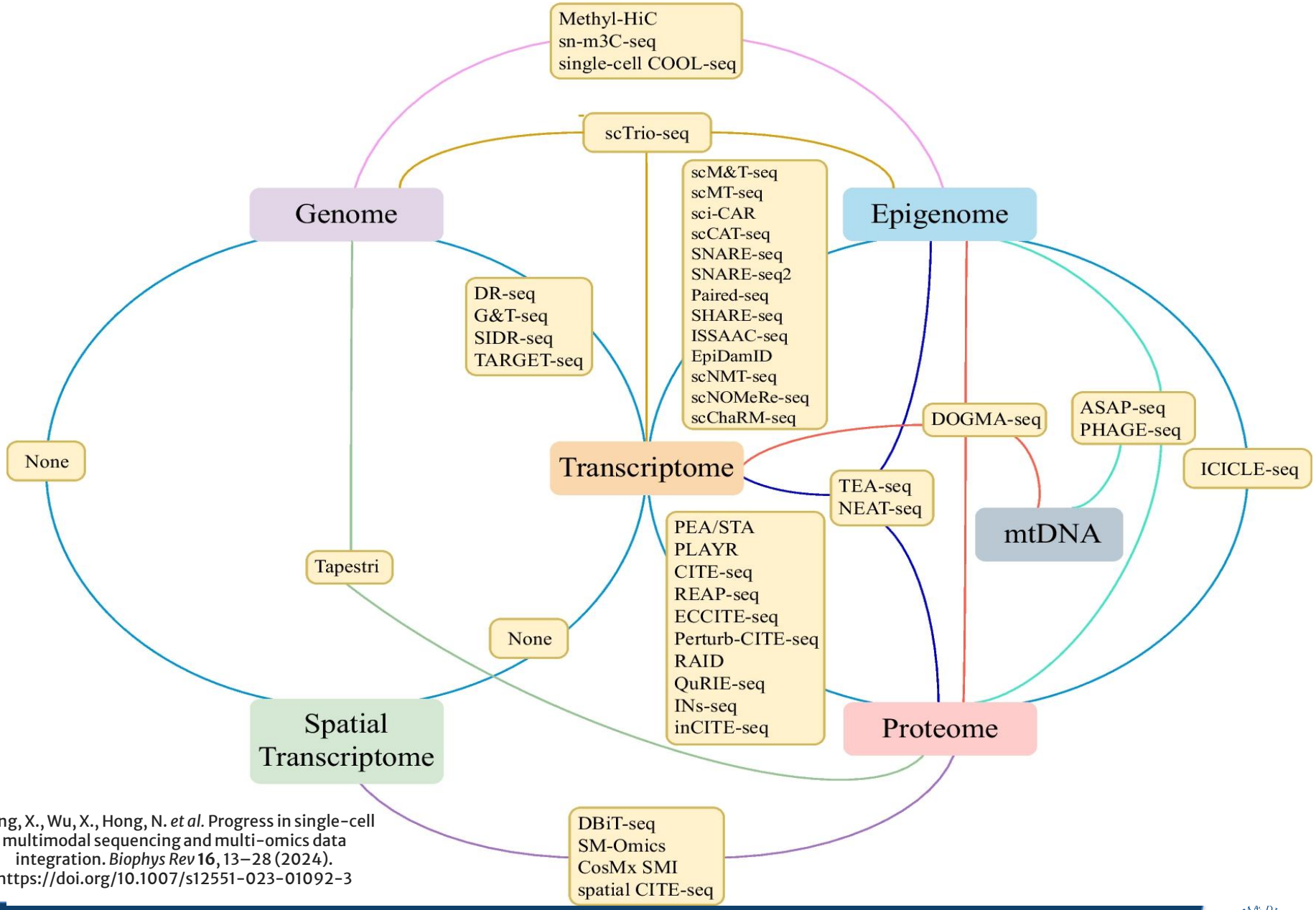
Different omics

#	Name	Information Obtained
1	Whole Genome Sequencing (WGS)	Complete DNA sequence of an organism's genome
2	RNA Sequencing (RNA-Seq)	Gene expression levels across the transcriptome
3	ChIP-Sequencing (ChIP-Seq)	Protein-DNA interactions identifying binding sites for transcription factors
4	Whole Exome Sequencing (WES)	Sequencing of protein-coding regions in the genome to identify genetic variants
5	Mass Spectrometry-based Proteomics	Identification and quantification of proteins in a sample
6	Methylation Sequencing	DNA methylation patterns across the genome (transcription repression)
7	ATAC-Sequencing (ATAC-Seq)	Chromatin accessibility, identifying open regions of chromatin
8	Hi-C or 3C	High-throughput Chromosome Conformation Capture and Chromosome Conformation Capture

Form different omics to sc multi-omics

Name	Reference	Information Obtained	Throughput
Single-Cell Triple Omics (scTrio-seq)	Bian, S., et al. (2018). Science.	DNA methylation, chromatin accessibility, RNA expression	Low
CITE-seq (Cellular Indexing of Transcriptomes and Epitopes by sequencing)	Stoeckius, M., et al. (2017). Nature Methods.	Protein abundance, RNA expression	High
REAP-seq (RNA Expression and Protein sequencing)	Peterson, V.M., et al. (2017). Nature Biotechnology.	Protein abundance, RNA expression	High
sciCAR-seq (Single-cell combinatorial indexing Chromatin Accessibility and mRNA sequencing)	Cao, J., et al. (2018). Science.	Chromatin accessibility, RNA expression	High
SNARE-seq (Single Nucleus ATAC-seq and RNA Expression sequencing)	Chen, S., et al. (2019). Nature Biotechnology.	Chromatin accessibility, RNA expression	High
G&T-seq (Genome and Transcriptome sequencing)	Macaulay, I.C., et al. (2015). Trends in Cell Biology.	DNA sequence, RNA expression	Low
scHi-C (single-cell Hi-C)	Ramani, Vijay, et al. Nature methods 14.3 (2017)	chromosome conformation capture techniques	High

Single-cell multimodal profiling methods

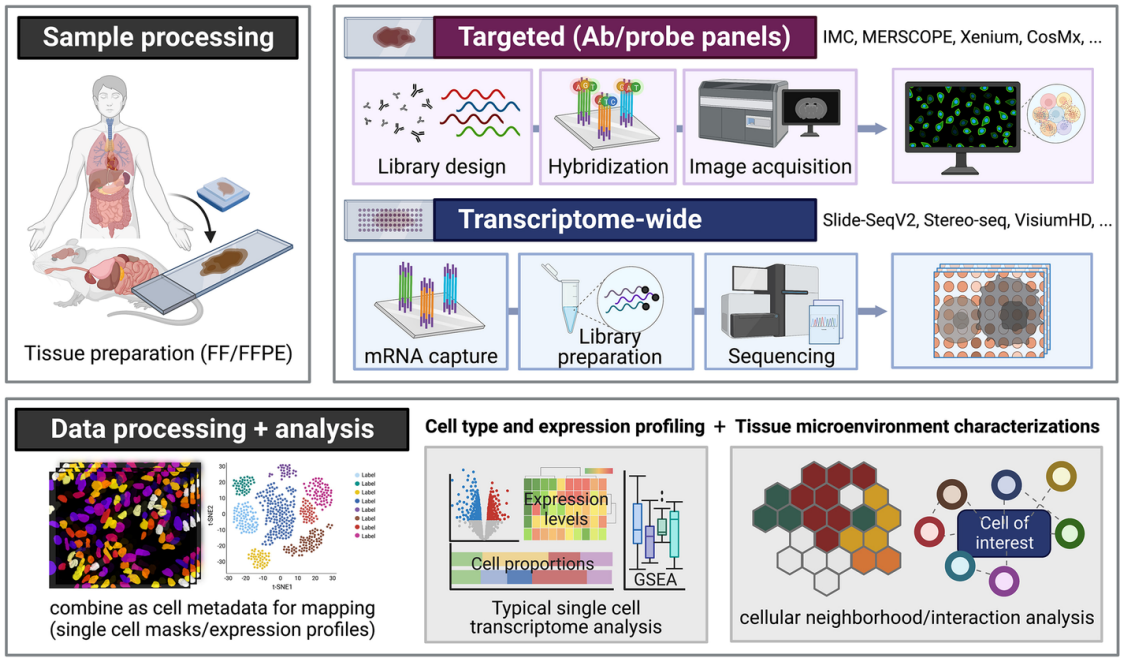


Wang, X., Wu, X., Hong, N. *et al*. Progress in single-cell multimodal sequencing and multi-omics data integration. *Biophys Rev* **16**, 13–28 (2024). https://doi.org/10.1007/s12551-023-01092-3

MEET IMAGES AND OMICS!

Spatially Resolved Single-Cell Data Analysis Tools

- Squidpy (Python),
- Seurat (R),
- Giotto (R)



Park, J., Kim, J., Lewy, T. *et al.* Spatial omics technologies at multimodal and single cell/subcellular level. *Genome Biol* **23**, 256 (2022). https://doi.org/10.1186/s13059-022-02824-6